DHANUSH M

Aspiring Software Development Engineer | MERN Backend | AI/ML Engineering | DSA

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Career Objective

Final-year Electronics and Communication Engineering student building toward a Software Development Engineer role with a focus on MERN backend systems, data structures and algorithms, and applied Python-based AI engineering. Experienced in shipping a complete edge-to-cloud AI system from model training through deployment, and in building IoT-integrated embedded systems.
Seeking an SDE or AI/ML engineering role where strong fundamentals and hands-on system-building can drive real product impact.

Education

Velalar College of Engineering and Technology — Aug 2023 – May 2027
Bachelor of Engineering (B.E.) in Electronics and Communication Engineering — *CGPA: 8.4/10*

Bharathi Matric Higher Secondary School, Vijayamangalam, Erode — Jun 2022 – Apr 2023
Higher Secondary (HSC) – Computer Science — *81%*

The Richmond Matric Higher Secondary School, Seenapuram, Erode — 2020 – 2021
Secondary School (SSLC)

Technical Skills

Programming Languages: Java, Python, C, SQL, JavaScript
Backend: MongoDB, Express.js, React.js, Node.js, REST APIs
Frontend: HTML, CSS, JavaScript
Databases: MySQL, Firebase Realtime Database, SQLite
AI/ML: YOLOv8, Model Training & Deployment
Embedded/IoT: ESP32, RFID, Biometric Authentication, Embedded C
Developer Tools: Git, GitHub, VS Code, IntelliJ IDEA, PyCharm, DataGrip, NVIDIA Build.com, WebStorm

Projects

RoadSense AI – Real-Time Road Damage Detection System — *Final Year Project*
Python, YOLOv8, Flask, Firebase, React.js, Raspberry Pi

- Designed and deployed a real-time road damage detection system using a custom-trained YOLOv8n model on a Raspberry Pi 4.
- Integrated GPS and MPU6050 IMU sensor fusion for automated damage localization.
- Built a complete edge-to-cloud pipeline using Firebase, SQLite, and a Flask REST API, connected to a React.js web dashboard for live monitoring and visualization.
- Trained and evaluated the model on the Rome/Sacrofano road damage dataset containing 2,009 images across 3 classes.

Integrated RFID-Enabled Digital Framework for Rabies Surveillance — *Mini Project*
ESP32, IoT, Firebase, Biometric Authentication

- Built an IoT-based healthcare monitoring system using ESP32, RFID (MFRC522), and R307 fingerprint biometric authentication for secure two-factor patient identification.
- Designed a real-time vaccination tracking pipeline using Firebase Realtime Database, replacing manual paper-based rabies surveillance records with centralized digital tracking.
- Implemented Embedded C firmware for RFID scanning, fingerprint verification, OLED status display, and Wi-Fi cloud synchronization end-to-end.

AI-Powered Train Traffic Control for Maximum Throughput — *SIH 2025 Prototype*
Python, AI, Optimization

- Worked on a prototype-level solution for Smart India Hackathon 2025 focused on improving railway traffic throughput and reducing scheduling conflicts.
- Contributed to idea design, problem analysis, and initial solution workflow using concepts related to AI-assisted decision making and optimization.

Innovative Passenger Tracking System in Public Transportation *SIH 2024 Prototype*
System Design, Public Transport Innovation

- Worked on a prototype and concept design for Smart India Hackathon 2024 aimed at improving passenger visibility and monitoring in public transportation systems.
- Contributed to problem identification, solution ideation, and workflow structuring for a technology-driven public transport use case.
- Strengthened understanding of solution architecture, collaborative ideation, and real-world problem solving through hackathon-based innovation.

Internships / Industrial Training

Edu-versity Feb 2026 – Apr 2026
Artificial Intelligence Intern *Remote*

- Artificial Intelligence internship focused on AI fundamentals, machine learning basics, and Python-based learning workflows.
- Developed understanding of data-driven reasoning, model-oriented thinking, and structured analytical problem solving.
- Gained practical exposure to real-world AI applications, technical learning environments, and foundational machine learning concepts.

Certifications

- **Udemy:** The Complete Full-Stack Web Development Bootcamp (62 hrs) – MERN Stack
- **NPTEL:** Programming in Java, Data Analytics with Python, Introduction to Machine Learning, Introduction to Large Language Models
- **GUVI (HCL):** React.js, JavaScript, HTML & CSS, Git, AI Fundamentals, Java Programming
- **Microsoft:** Fundamental AI Concepts
- **Udemy:** Git & GitHub, C Programming
- **Scaler:** Python Programming, Logic Building
- **Aptiwings:** Aptitude, Problem Solving

Achievements

- **Smart India Hackathon (SIH) 2024:** Selected at the inter-college idea evaluation level, demonstrating problem-solving and solution ideation skills.
- **Smart India Hackathon (SIH) 2025:** Selected at the inter-college idea evaluation level, demonstrating problem-solving and solution ideation skills.

Leadership & Extra-Curricular Activities

- **IEEE Student Branch Chairperson:** Led student-level technical activities, coordinated professional and technical initiatives, and contributed to knowledge sharing and community engagement.

Soft Skills

Problem Solving | Communication | Teamwork | Adaptability | Time Management | Leadership | Analytical Thinking

Areas of Interest

MERN Backend Development | Artificial Intelligence | Machine Learning